DAILYN DESPRADEL RUMALDO

347-357-4148 ddesprad@andrew.cmu.edu linkedin.com/in/dailyn-despradel dailyndespradel.github.io

Education

Carnegie Mellon University (CMU), Pittsburgh, PA **GPA: 3.77/4.00**
Ph.D. in Mechanical Engineering,Human-Computer Interaction Focus Defense: Oct 2026 — Graduation: Dec 2026

New York Institute of Technology (NYIT), New York City, NY **GPA: 3.66/4.00**
M.S. in Electrical and Computer Engineering Graduation: May 2021

Mercy College, Dobbs Ferry, NY **GPA: 3.71/4.00**
B.S. in Biology Graduation: May 2018

PhD Research Experience

NeuroMechatronics Lab (Dr. Douglas Weber), Mechanical Engineering, CMU **August 2021 – Present**
Research Lead / PhD Thesis: *Adaptive Extended Reality (XR) Interactions: Understanding and Enhancing User Experience*

- **Adaptive XR Systems:** Develop adaptive and intent-aware XR interactions using **Surface Electromyography (sEMG), gaze, motion, and hand kinematics** for real-time digital interaction.
- **Machine Learning + Real-Time Adaptation:** Build user-data-driven machine learning pipelines for force- and intent-aware XR interaction adaptation and object manipulation.
- **XR Development:** Design immersive 2D/3D XR experiences in **Unreal Engine** and **Unity**, integrating eye tracking, hand tracking, and multimodal sensing.
- **3D Prototyping:** Prototype XR interaction hardware and 3D assets using **Blender**, **Autodesk Fusion 360**, and rapid prototyping workflows.
- **Human Movement + Object Manipulation:** Investigate neuromechanics of object manipulation across physical and virtual environments using sEMG, motion capture, and kinematic analysis.
- **Industry Collaboration:** Collaborate with **Meta** on myoelectric interaction prototypes and accessible XR interaction techniques for individuals with motor impairments.

Work Experience

Research Scientist Intern, Meta Inc. **Oct 2025 - March 2026**
User Experience and Robotics Systems

- Integrated physiological sensing and adaptive interaction techniques to support intent-aware robotic systems and improve human–robot interaction.
- Developed XR interaction and robotics prototyping workflows using **Unreal Engine**, **Unreal Engine Blueprints**, **Python**, and **SolidWorks**.

Vision Pro Group (VPG) Intern, Apple Inc. **April 2025 - September 2025**
Vision Pro Ergonomics / Product Design Analyst Intern

- Conducted ergonomic assessments and data-driven analyses of Vision Pro hardware to optimize comfort, wearability, and user experience across diverse user populations.
- Utilized **Python**, **NX**, and **3D scanning technologies** to analyze hardware fit, user variability, and product design considerations.

Skills

Research Areas: Human-Computer Interaction (HCI), Extended Reality (VR/AR/MR), Adaptive and Intent-Aware Interfaces, Physiological Computing, Neuromechanics, Human Motion Analysis, Object Manipulation, Accessible Interaction Design
Programming: Python, C#, Unreal Engine Blueprints
XR + Development Tools: Unity, Unreal Engine, WebXR, Blender, Autodesk Fusion 360, SolidWorks, NX
Machine Learning + Data Analysis: Real-Time Signal Processing, Machine Learning Pipelines, EMG Analysis, Motion Capture Analysis, Multimodal Data Analysis
Hardware + Sensing: sEMG, Eye Tracking, Hand Tracking, Motion Capture (Vicon), 3D Scanning
Languages: Spanish (Native), English (Professional Fluency)

Fellowships

- GEM Fellow, CMU, Mechanical Engineering, 2021 - Present
- McNair Scholar, Mercy College, Biology, 2018

Publications

- **D. Despradel**, et al. "A Wearable Myoelectric Sensor Enables Human–Machine Interaction for People with Tetraplegia." *Science Robotics*, Under Review.

Master's Research Experience

Electrical and Computer Engineering, NYIT (Advisor: Dr. N. Sertac Artan) **Aug 2019 – May 2021**
Research Lead: *Virtual Reality and EMG-Based Interaction for Prosthetic Rehabilitation*

- Developed VR rehabilitation interactions using the **Oculus Quest 2** and **Myo Armband (EMG)** for gesture-based virtual arm control.
- Designed and evaluated real-time control of four virtual hand gestures, including fist, open hand, wrist flexion, and wrist extension.
- Analyzed latency and responsiveness between EMG sensing and VR interaction pipelines to improve interaction performance and usability.
- Investigated accessible and gesture-driven interaction techniques for future prosthetic rehabilitation and assistive XR applications.

Graduate Coursework

In-Person Learning: Extended Reality interactions by the Human-Computer Interaction CMU program, Crafting Software, Machine Learning, Human Anatomy and Physiology, Neural Engineering Laboratory, Signals and Systems, Communication Theory
Online Learning (Coursera and edX): Principles of UX/UI Design, 3D Models in Virtual Reality, 3D Models for Virtual Reality, Interactive Computer Graphics, Developing AR/VR/MR/XR Apps with WebXR, Unity and Unreal

Poster Presentations

- **D. Despradel**, et al. "Enabling Advanced Interactions through Closed-Loop Control of Motor Unit Activity After Tetraplegia." *ACM Symposium on User Interface Software and Technology (UIST)*, Pittsburgh, PA, 2024.
- **D. Despradel**, et al. "Closed-Loop Control of Motor Unit Activity After Tetraplegia for Human-Computer Interaction." *Society for Neuroscience Annual Meeting*, Chicago, IL, 2024.
- **D. Despradel**, et al. "Real-Time Detection and Decoding of Motor Unit Activity Using a Wearable Neuromotor Interface in a Person with Motor Complete Tetraplegia." *Neural Control of Movement*, Dubrovnik, Croatia, 2024.
- **D. Despradel**, et al. "Exploring the Neuromechanics of Feedback-Influenced Object Manipulation in Virtual and Physical Environments." *Society for Neuroscience Annual Meeting*, Washington, DC, 2023.
- **D. Despradel**, et al. "Understanding the Neuromechanics of Haptic-Free Object Manipulation in Virtual and Physical Environments." *Society for Neuroscience Annual Meeting*, San Diego, CA, 2022.

Outreach & Leadership

- **Black History Month CMU MechE DEI VR Event** – Demonstrated immersive VR technologies and interactive experiences for community engagement and STEM outreach, 2024.
- **Biomechanics Day Outreach Event** – Taught high school and middle school students principles of biomechanics through animated character motion, XR interaction, and digital human movement analysis, 2024.
- **Biomechanics Day Outreach Event** – Led hands-on demonstrations on hand rigging, animation, and biomechanics for K–12 STEM outreach participants, 2023.
- **MEGSO (Mechanical Engineering Graduate Student Organization)** – Mentorship Chair, organizing mentorship initiatives and graduate student community engagement, 2023–2024.